

ITA05-ACTIVITY

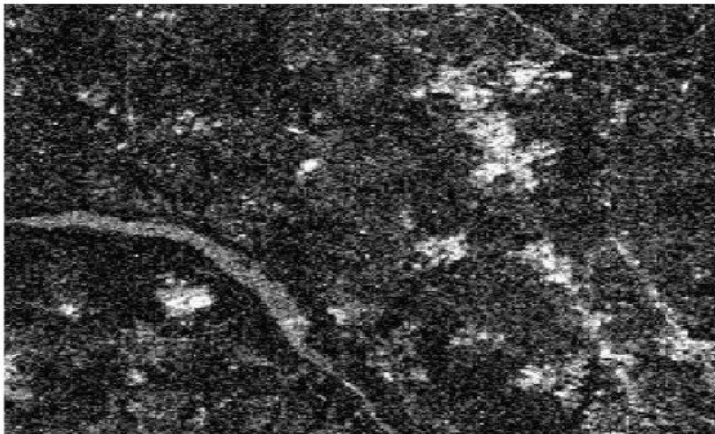
SCENARIO BASED ACTIVITY

Mention the techniques used with reason:

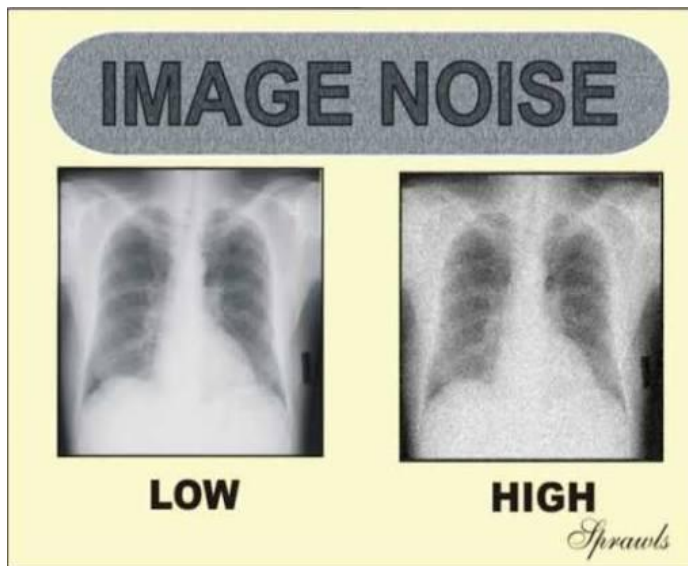
1. A CCTV image contains **salt-and-pepper noise** due to transmission errors.



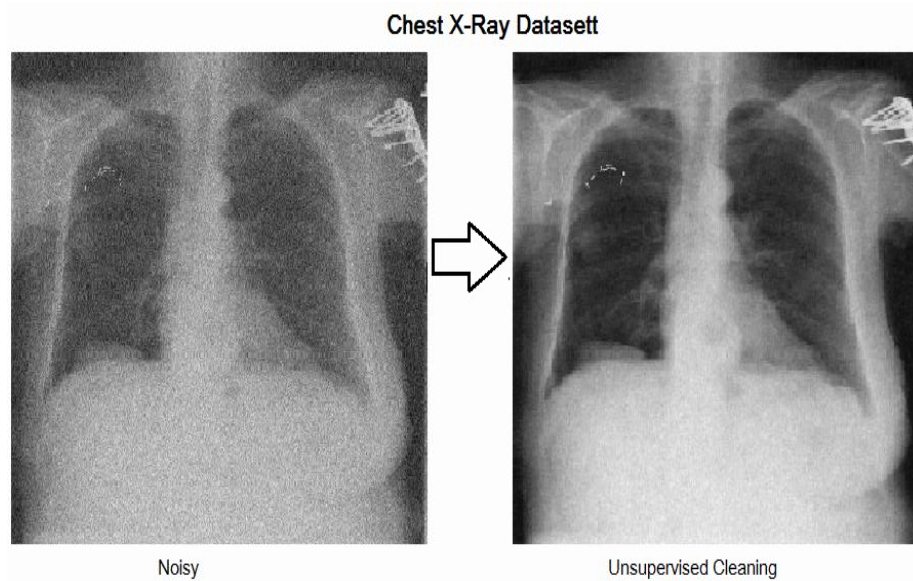
2. A satellite image is corrupted by **Gaussian noise** from sensors.



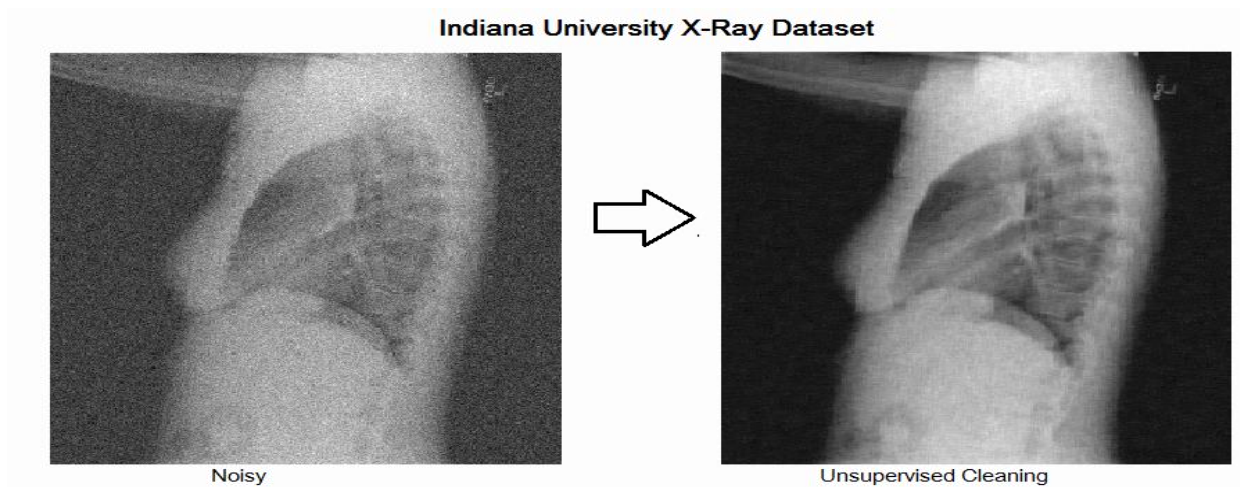
3. A medical X-ray image has random noise but fine edges must be preserved.



4. A scanned document has **black and white dots** scattered randomly.



5. A low-light image shows heavy grain without clear boundaries.



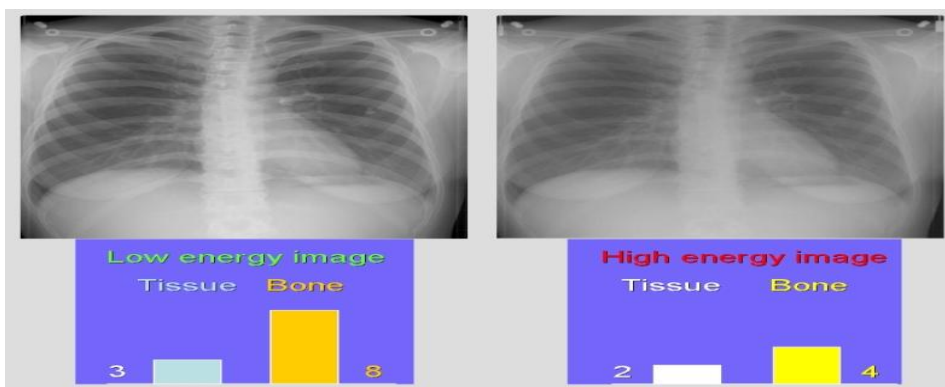
6. A foggy road image has **very low contrast**.



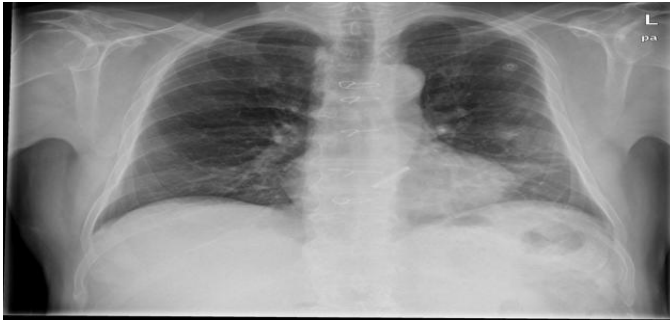
7. An underwater image appears dull and washed out.



8. A chest X-ray shows poor visibility of internal structures.



9. A night-time traffic image lacks intensity variation.



10. A grayscale image uses only a **narrow intensity range**.



Activity - 2

1. Scenario:- CCTV image contains salt and pepper noise due to transmission errors

Technique used:- Median Filter

Reason:- Removes salt and pepper (impulse) noise effectively while preserving image edges

2. Scenario:- Satellite image is corrupted by gaussian noise from sensors

Technique used:- Gaussian filter

Reason:- Smooths gaussian noise by averaging neighbouring pixel values

3. Scenario:- Medical x-ray image has random noise but fine edges must be preserved.

Techniques used:- Bilateral filter

Reason:- Reduces noise while preserving important edges and fine details

4. Scenario:- Scanned document has black and white dots scattered randomly.

Technique used:- Median Filter

Reason:- Removes black and white dots without damaging text (salt and pepper noise)

5. Scenario:- Low light image shows heavy grain without clear boundaries

Technique Used:- Gaussian filter

Reason:- Reduces grainy noise and produces a smoother image.

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6. Scenario:- Foggy road image has very low contrast.

Techniques Used:- Histogram Equalisation

Reason:- Enhances image contrast and improves visibility in foggy conditions

7. Scenario:- underwater image appears dull and washed

Techniques used:- contrast stretching

Reason:- Improves brightness and contrast, making underwater details more visible

8. Scenario:- Chest X-ray shows poor visibility of internal structures

Techniques used:- Adaptive Histogram Equalisation (CLAHE)

Reason:- Enhances local contrast and reveals internal structures clearly

9. Scenario:- Night time traffic image lacks intensity variation

Techniques used:- Histogram Equalisation

Reason:- Increases intensity variation and improves visibility in dark images

10. Scenario:- Grayscale image has only a narrow intensity range

Techniques used:- contrast stretching

Reason:- Expands the intensity range, resulting in better overall contrast.